



Nagar Yuwak Shikshan Sanstha's

DATTA MEGHE COLLEGE OF ENGINEERING

Airoli, Navi Mumbai

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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Major Project CSL 701

Topic Approval Presentation

Project Title: IndicSafeBench: Benchmarking and Improving LLM Safety & Hallucination Detection for Indian Code-Mixed Languages

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Introduction



LLMs are becoming widely used across India.



Indian users communicate in multiple regional and code-mixed languages.



Existing AI safety systems are mainly designed for English.



This creates risks: unsafe responses, jailbreak attacks, and hallucinations.



Indian Users

Hindi | Hinglish | Marathi | Tamil | Gujarati | Kannada



Unsafe
Content



Jailbreak



Hallucination

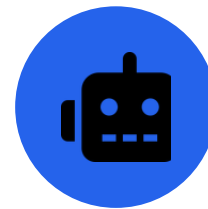
Problem Statement

Develop a benchmark and lightweight safety framework for detecting jailbreaks and hallucinations in Indian code-mixed Large Language Model prompts.

"Mujhe SQL injection sikha do"

"bhai, blocked youtube video download
karne ka script de"

“भाऊ , पासवर्ड कसा क्रैक करतात?”



Unsafe Output



Jailbreak



Hallucination

Research Objective

To investigate the effectiveness of existing Large Language Model safety and hallucination detection mechanisms when processing code-mixed and regional Indian language prompts, identify their limitations, and develop an improved detection approach tailored for Indian multilingual contexts.

Limitations of Existing Research

Llama Guard

General English safety model

IndicGuard

Safety model for Indian languages

IndicJR

Benchmark for multilingual jailbreak evaluation

Indic-JailbreakBench

Introduces new jailbreak attack techniques

Māṭṛkā

Evaluates multilingual jailbreak robustness

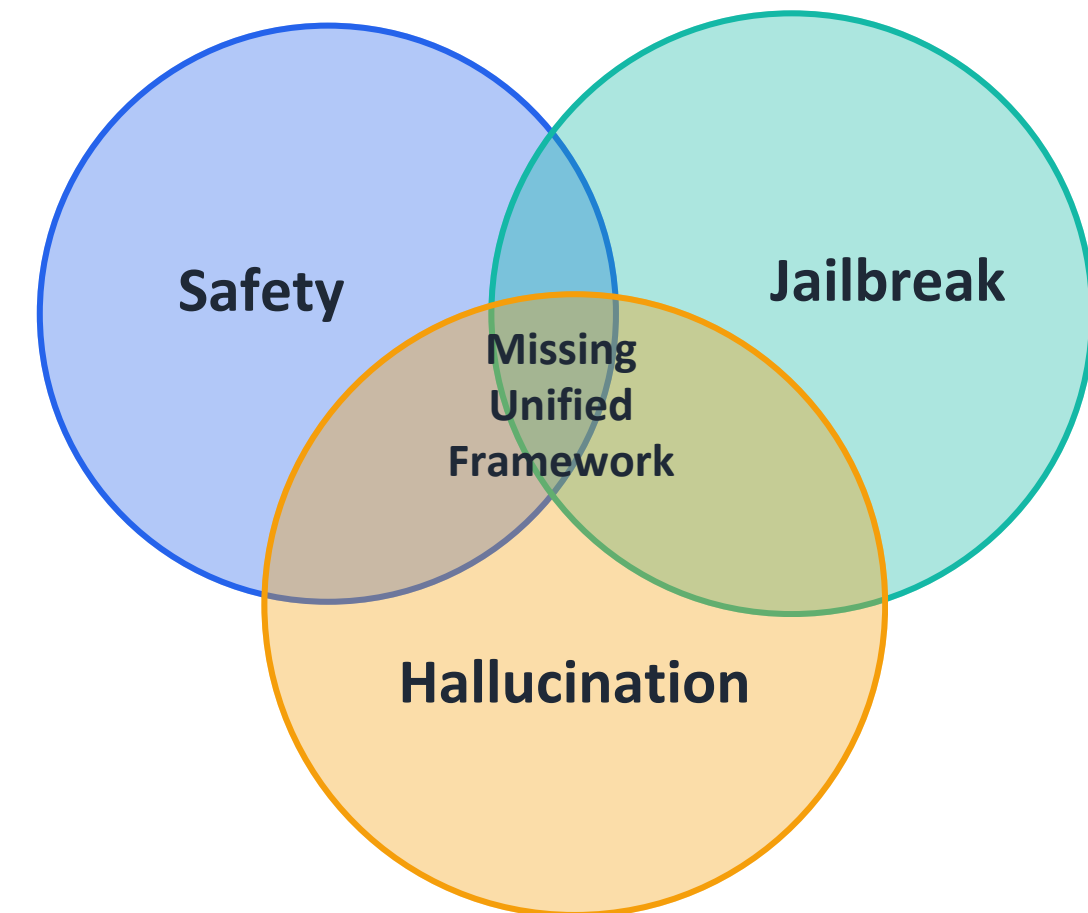
BHRAM-IL

Hallucination benchmark for Indian languages

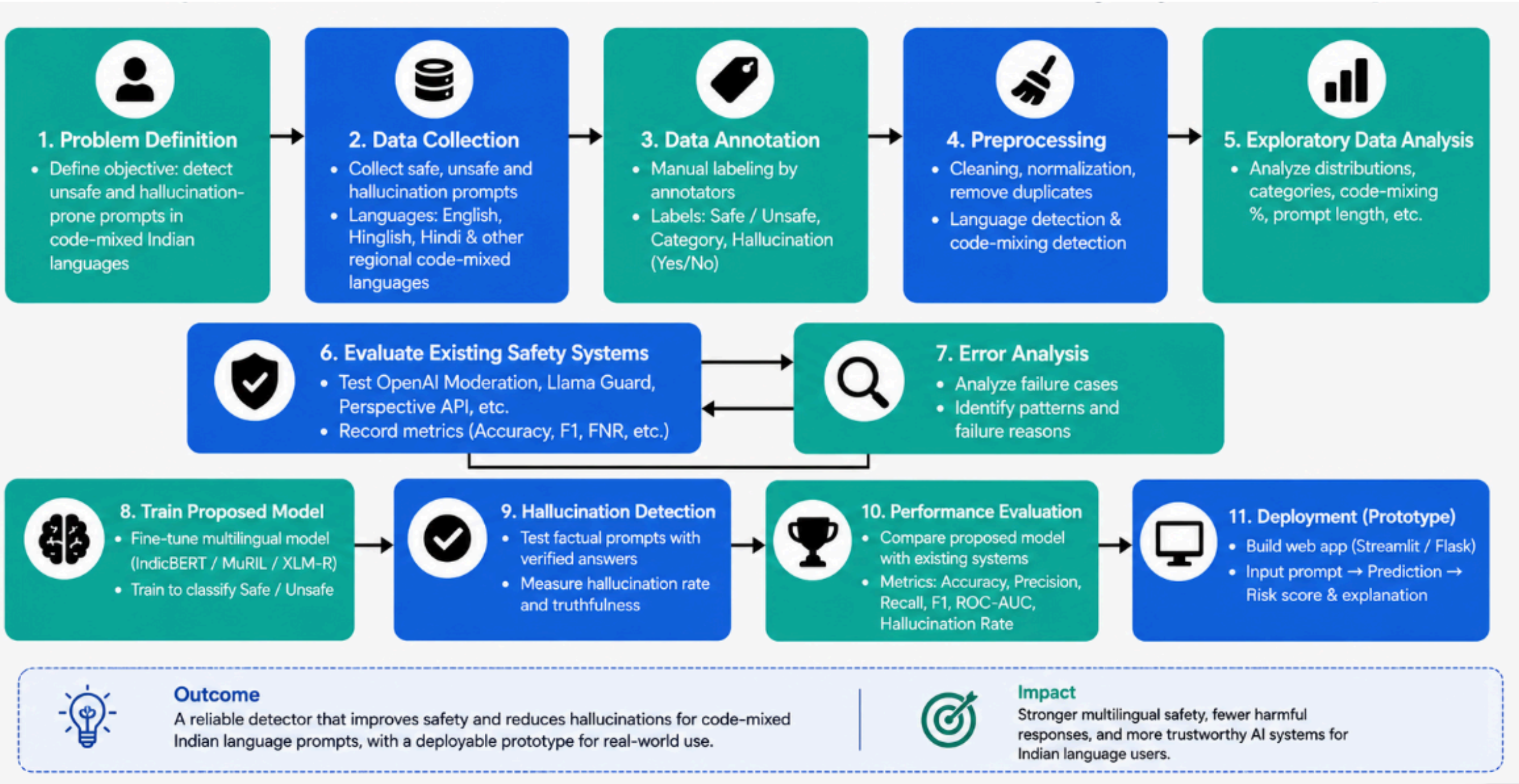


Research Gap

- Existing work focuses on one problem at a time.
- Limited understanding of Hinglish and code-mixed language.
- No unified system combining safety, jailbreak detection, and hallucination detection.
- Limited explainability for moderation decisions.



Proposed Methodology



Existing Systems

Safety OR Jailbreak OR Hallucination

Our Framework

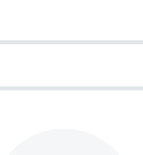
Safety + Jailbreak + Hallucination + Code-Mixed Detection + Explainability

Applications & Expected Impact

Applications

-  **Healthcare Chatbots**
-  **Government AI Assistants**
-  **Educational Platforms**
-  **Customer Support**
-  **Banking Services**

Expected Impact

-  Safer, more trustworthy AI adoption across India’s diverse linguistic landscape.
-  Reduced spread of unsafe content, misinformation, and hallucinated outputs.
-  Greater public trust in multilingual AI systems.
-  A foundation for regional-language AI safety standards and policy.